Mapping "Cult": A Plain-Language Walkthrough of the Geometric Analysis

Working report – draft base material for the Results chapter

Prepared for Célestin Meunier's thesis project

Covers run `20260907-002832`, generated 2026-09-07

How to read this document. This is not a finished Results chapter. It is a neutral, plain-language walkthrough of (1) what has been built so far, and (2) what the geometric-analysis toolkit found when it was run end-to-end on the current dataset. Numbers here are reported, not interpreted – turning them into an argument about what "cult" actually means across sources is the next, separate step, to be done by the researcher. Anywhere a number could be misread as stronger evidence than it is, that caveat is kept in, not smoothed over.

Contents

1 Where the project stands

The thesis asks a simple-sounding question with a genuinely open answer: when different groups of people – scholars, a French government agency, and ordinary interviewees – talk about "cults," are they talking about the same thing? Rather than starting from one fixed definition and checking who fits it, the project tries to *map* how the idea of "cult" is actually used across three very different sources, and see where those maps agree, disagree, or fail to overlap at all.

Getting to a point where that question can be asked geometrically took several distinct stages of work. This section summarises them in order.

1.1 Stage 1 – Building three text corpora

Three separate bodies of text were assembled, each representing a different **epistemology** (a fancy word for "a way of knowing something" – here, a distinct source of authority on what counts as a cult):

- **Scholarly literature**: 57 books, book chapters, and journal articles on cults and New Religious Movements, selected through a reproducible search-and-screening protocol against the OpenAlex academic database (keyword queries, field/subfield exclusions to remove unrelated senses of "cult" such as ancient ritual practice or pop-fandom, a 2001–2026 date window, and citation-based ranking).
- **MIVILUDES**: the French government's inter-ministerial mission against sectarian abuses. Two official documents – their 2022–2024 activity report and their public "how to identify a sectarian drift" reference page, which lists 17 official criteria.
- **Interviews**: 26 semi-structured interviews with ordinary people (20 in person, 6 via Instagram direct message; 21 in English, 5 in French), opening with a single free-association prompt ("what comes to mind when you hear the word cult?").

1.2 Stage 2 – Turning raw text into structured "expressions"

Each document (once converted to clean text, with a scanned-page OCR fallback where needed) was fed in chunks to a locally-run language model (`qwen3:4b`), instructed to pull out **expressions**: self-contained statements that describe, define, dispute, or otherwise engage with what makes something cult-like. Each expression was tagged with *how* it makes its claim (a direct statement, a quotation, a definition, a question, . . .), *how certain* it is (asserted, qualified, contested, negated, speculative), and *who* it is attributed to. This produced 39,236 expressions from the literature, 914 from MIVILUDES, and 230 from the interviews.

1.3 Stage 3 – Building three "reference vocabularies"

To have something stable to compare the three corpora against, three background word-lists were built (details of what these mean, and why three rather than one, are in Section 2):

- **Concept backbone** (3,000 terms): topic-neutral, general English concepts drawn purely from WordNet's own structure, with no reference to cults at all.
- **Structural concepts** (1,500 terms): the opposite extreme – generic, structural/relational vocabulary (e.g. "authority," "control," "isolation") mined directly out of the corpora's own expression text.
- **ConceptNet-generalised concepts** (195 terms): a middle ground, generalising the structural-concepts list outward through ConceptNet's associative word-graph, then hand-reviewed one candidate at a time to keep only genuinely on-topic terms.

1.4 Stage 4 – Extracting "emergent entities"

Every expression is also tagged with the named things it mentions – "Scientology," "charismatic leader," "Heaven's Gate," and so on. Pooling and normalising these tags across all three corpora, and keeping only ones mentioned at least 3 times, produced 3,251 **emergent entities**: named things the corpora themselves talk about, as opposed to a claim a source makes or a neutral reference term.

1.5 Stage 5 – One shared geometric space

All of the above – expressions, the 17 official MIVILUDES criteria, the three reference vocabularies, and the emergent entities, eight sources in total – were converted into numbers (**embedded**, see Section 2) with the same model, then pooled into a single **shared space** so that a point from any one source can be compared directly to a point from any other. Filtering out exact duplicate expressions and very short fragments left 44,520 points, compressed from the embedding model's native 1,024 dimensions down to 394 while keeping 95% of the original variation.

1.6 Stage 6 – The interview "first impression" layer

Interview responses to the opening prompt were often too short to survive the length filter used everywhere else (e.g. "AI cult," "Illuminati!"), even though a short, spontaneous first answer is exactly what that part of the interview was designed to capture. Rather than loosening the filter for everyone (which would have reshuffled the whole space), each of the 26 interviews' opening answer was manually checked against the transcript and placed into the same coordinate system as a separate layer. 25 of 26 resolved this way; one interview had no qualifying opening claim at all.

1.7 Stage 7 – Building and running the analysis toolkit

A reusable toolkit of nine analysis modules was then built and run end-to-end against the current 44,520-point space (this is the run covered by this document, id `20260907-002832`, git commit `40f94fbfd932`). What each module does and found is the subject of Section 4.

1.8 Stage 8 – Where things stand now

The toolkit produces tables, figures, and a neutral draft summary – *not* a finished analysis. No run's output has yet been interpreted or written into the thesis's Results chapter. This document is the bridge: an explained, illustrated version of what the toolkit currently shows, meant to be read, argued with, and turned into the actual Results prose.

2 Key ideas, explained in plain language

This section explains, without assuming prior background, every technical term used later in this document. Skip it if you are already comfortable with embeddings and dimensionality reduction.

2.1 Embeddings and vector spaces

An **embedding** is what you get when a piece of text (a sentence, a phrase, a single word) is fed through a trained language model and converted into a long list of numbers – a **vector**. Here every vector has 1,024 numbers. The key property that makes this useful: *texts with similar meaning end up as nearby lists of numbers.* "A group led by a controlling figure" and "an authoritarian leader dominates the members" would land close together; "a group led by a controlling figure" and "the recipe calls for two eggs" would land far apart. A collection of such vectors is called a **vector space** or **embedding space**, and because "nearby in the list of numbers" tracks "similar in meaning," ordinary geometric ideas – distance, direction, clustering – become tools for studying meaning.

2.2 Distance and similarity

Two ways of measuring "how far apart" two points are, both used throughout:

- **Euclidean distance**: ordinary straight-line distance, the same idea as distance on a map, just in many more than two dimensions. Smaller = more similar.
- **Cosine similarity**: instead of measuring the gap between two points, this measures whether they point in the *same direction* from the origin, ignoring how long each vector is. It ranges from -1 (opposite directions) to 1 (exactly the same direction), with 0 meaning unrelated/perpendicular directions.

The two measures do not always agree, and both are reported side by side rather than picking one as "the" answer.

2.3 Centroid

The **centroid** of a group of points is just their average position – add up all their vectors and divide by how many there are. It is a stand-in for "the centre of gravity" of, say, everything the literature corpus says, so that "how far apart are literature and MIVILUDES, roughly speaking" can be answered with one number (the distance between their two centroids) instead of comparing every pair of points individually.

2.4 Standardization and PCA (dimensionality reduction)

Before the eight sources were pooled into one space, every dimension of every vector was **standardized** – rescaled to have a comparable spread – so that no single source could dominate the result just because its raw numbers happened to be larger, given how different academic prose, government French, casual interview speech, and bare dictionary glosses are as writing styles.

PCA (Principal Component Analysis) is then a way of compressing that giant 1,024-number description of each point down to a smaller number of dimensions that still captures most of the original information, by finding the directions along which the data actually varies the most and discarding the directions that carry almost no information. Here, the smallest number of dimensions that still keeps 95% of the original variation is 394 – a real compression, but not a drastic one (the curve is gradual: 100 dimensions alone would only keep 61% of the variation). Every later distance and similarity number in this document is computed in this 394-dimensional space, not the original 1,024.

2.5 UMAP and 2-D pictures

394 dimensions cannot be drawn on a page. **UMAP** is a separate technique used purely for *visualisation*: given a high-dimensional set of points, it produces a 2-D layout that tries to preserve which points were close together and which were far apart, so the resulting scatter plot is a readable (if approximate and sometimes distorted) picture of the real structure. A plain 2-D **PCA slice** (just the first two principal components from the main reduction above) is shown alongside UMAP for the same reason two different projections of a 3-D object can each tell you something the other misses – and so that figures stay comparable to each other under the same fixed linear projection. Neither picture is itself evidence; the actual numbers come from the full 394-dimensional space. Pictures are a reading aid.

2.6 k-nearest-neighbour composition

For a given point, its **k nearest neighbours** are simply the k closest other points in the space (this document uses $k = 10$ and $k = 20$). If a literature point's 10 nearest neighbours are, say, 4 from literature, 3 from MIVILUDES, and 3 from interviews, that is a very mixed neighbourhood; if all 10 are from literature, that source is sitting in a much more self-contained region of the space. Averaging this composition over many points from a given source is one concrete way to ask "how much do these three sources actually overlap, versus sitting in their own separate regions?"

2.7 Silhouette score

A single number, in principle between -1 and 1, that scores how well a proposed grouping (here: the three expression sources) matches the actual geometric clustering of the data – roughly, "on average, is each point closer to others in its own group than to points in other groups?" A score near 0 means the proposed groups do not correspond to distinct geometric clusters; a score near 1 would mean they line up almost perfectly with the actual shape of the data.

2.8 Bootstrap resampling and confidence intervals

Several figures below are reported as a mean plus a **95% confidence interval**, e.g. "mean 0.32, 95% CI [0.28, 0.35]." This comes from **bootstrap resampling**: repeating a measurement many times (100 times here) on randomly resampled subsets of the data, then looking at how much the answer varies across those repetitions. A narrow interval means the measurement is stable regardless of exactly which points happened to be sampled; a wide one means the number is noisier than it looks at first glance.

2.9 "Point role" versus "source dataset"

Every point in the shared space carries two different labels that answer two different questions:

- **source_dataset** – which of the eight underlying datasets a point came from (literature, MIVILUDES, interviews, MIVILUDES's 17 criteria, the three reference vocabularies, or the emergent entities). The finest-grained label.
- **point_role** – a coarser grouping into three kinds of thing: an **expression** (something a source actually said or claimed), a **reference** (one of the three background vocabularies – not a claim, just a backdrop), or an **emergent** entity (a named thing the corpora talk about).

Figures are generally shown twice, coloured once by each label, since they answer different questions ("which specific source is this from" vs. "what kind of thing is this, regardless of source").

2.10 "Equal-size-controlled" and "equal-n" comparisons

The three sources are wildly different in size (literature: 35,621 points; MIVILUDES: 732; interviews: 204), and the three reference vocabularies are too (3,000 / 1,500 / 195 terms). A raw comparison – "which reference term is nearest, on average" – would be biased simply by pool size: a vocabulary with 3,000 candidates has more chances of having something coincidentally nearby than one with 195. An **equal-size-controlled** comparison down-samples every set to match the smallest one before comparing, so the comparison reflects actual closeness rather than candidate-pool size. This controls for *one* specific bias (size) – it is not evidence that the vocabularies are otherwise interchangeable, since they still differ in where they came from, how they were curated, and what they are meant to capture.

3 The eight datasets pooled into the shared space

Source	Points	Role	What it is
literature	35,621	expression	Claims/definitions extracted from 57 scholarly works
miviludes	732	expression	Claims extracted from 2 MIVILUDES documents (English translation embedded, French kept as label)
interviews	204	expression	Claims extracted from 26 interview transcripts
miviludes_criteria	17	expression	MIVILUDES's own official 17 sectarian-drift criteria (French, embedded as-is)
concept_backbone	3,000	reference	Topic-neutral general English concepts (WordNet-derived)
structural_concepts	1,500	reference	Generic structural/relational vocabulary mined from the corpora's own text
conceptnet_concepts	195	reference	Hand-reviewed generalisation of structural_concepts via ConceptNet
emergent_entities	3,251	emergent	Named things (people, groups, concepts) the corpora mention $\geq$3 times

Table 1: The eight sources pooled into the 44,520-point, 394-dimensional shared space.

A fourth, separate layer – the 25 manually-reviewed **interview prototypes** (`point_role="prototype"`) – sits in the same coordinate system but is *not* part of the 44,520-point count or the PCA fit itself (Stage 6 above).

4 What the analysis toolkit found

The toolkit has nine modules. They were all run against the current space in run `20260907-002832`. Each subsection below covers one module: what it does, in plain language, and what it found this run.

4.1 Sanity check: can interview order stand in for what people think of first?

Module: `audit_free_listing_rank`

Every extracted interview expression carries a `response_rank`: its position in the order items were pulled out of the transcript. It is tempting to treat "rank 1" as "the very first thing this person thought of" – a genuine, useful signal about spontaneous association. This module checked that assumption directly, and it does **not** hold:

Verdict: not usable. `response_rank` is an artefact of extraction order through the *whole transcript*, not a ranked list of associations. Even the narrowest, most generous reading – "does rank 1 land on the participant's own first claim, rather than e.g. the interviewer's question text?" – holds for only 11 of 26 interviews (42.3%). This also matches how the interview was actually structured: participants were asked for *one* example, then asked follow-up questions about *that same* example, rather than asked to list several items in order (which is what a genuine "free-listing" task would look like).

Consequence: nowhere in this document, or in the toolkit's other modules, is `response_rank` used to make any claim about what interviewees thought of "first" in a cognitively meaningful sense. The manually-reviewed interview-prototype layer (Section 4.5) is used instead, precisely because it does not depend on this broken assumption.

4.2 Global structure: how far apart are the sources, overall?

Module: `analyze_global_structure`

This module computes the centroid of each of the eight sources and measures the distance between every pair – literature's "average position" versus MIVILUDES's, versus interviews', and so on.

Source A	Source B	Euclidean distance	Cosine similarity
literature	miviludes	7.18	−0.04
literature	interviews	9.97	0.04
miviludes	interviews	12.25	−0.06
literature	miviludes_criteria	13.94	−0.26
miviludes	miviludes_criteria	10.89	0.57
interviews	miviludes_criteria	17.64	−0.15

Table 2: Selected pairwise centroid distances between the three expression sources (and MIVILUDES's own criteria list). Full 8×8 matrix in `pairwise_source_centroid_matrix.csv`.

In plain terms: on average, literature and MIVILUDES sit closest together of the three expression sources (7.18), interviews sit somewhat further from both (9.97 from literature, 12.25

from MIVILUDES). MIVILUDES's own 17 official criteria sit closer to MIVILUDES's general text (10.89, with a positive cosine similarity of 0.57 – the only clearly positive pairing in this table) than to literature or interviews, which makes sense: the criteria are MIVILUDES's own vocabulary. None of these numbers alone say whether a gap is "large" or "small" in a meaningful sense – that requires a baseline, which is exactly what the cluster-structure module (Section 4.3) supplies.

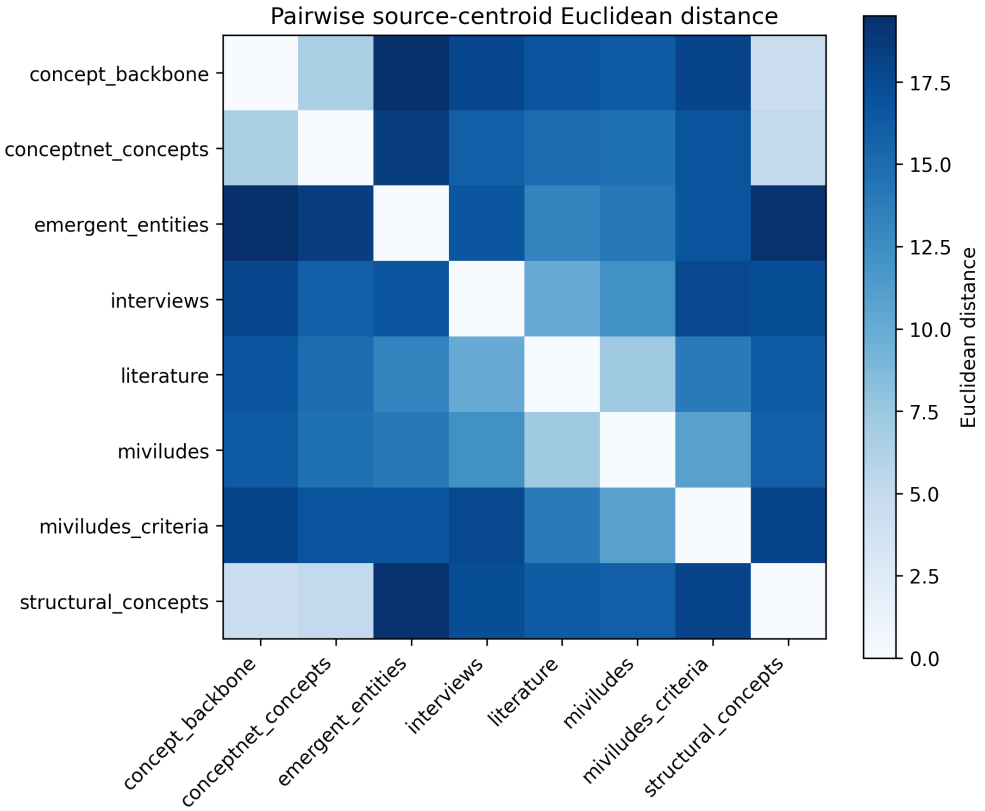

Figure 1: Heatmap of pairwise centroid distances across all eight sources. Darker/lighter cells indicate closer/further centroids; see the module's own colour scale.

The same module also measured how far each of the three reference vocabularies and the emergent-entities set sit from the combined expression centroid (literature+MIVILUDES+interviews together, computed three different ways – see the CSV for the "full" vs. "reduced_literature" vs. "equal_weight" variants, which are never merged into one number). Using the most direct ("full") variant: concept_backbone 16.70, structural_concepts 16.18, conceptnet_concepts 14.92, emergent_entities 13.20. The ordering is consistent with how each vocabulary was built: emergent entities (drawn directly from what the corpora mention) sit closest, the topic-neutral concept backbone sits furthest, and the two vocabularies built as deliberate middle grounds (structural concepts, ConceptNet-generalised concepts) land in between – with the ConceptNet-generalised set, hand-picked for domain relevance, closer than the corpus-mined but unreviewed structural-concepts list. This is a centroid-level comparison only (not adjusted for the three vocabularies' very different sizes – 3,000/1,500/195 terms); a separate, size-controlled comparison exists alongside it for exactly that reason and is not to be averaged with this one.

4.3 Cluster structure: do the three sources actually separate?

Module: `analyze_cluster_structure`

Centroid distances (previous section) say how far apart the *average* positions are, but not whether the sources form genuinely separate regions or heavily overlapping clouds. This module answers that with k-nearest-neighbour composition and a silhouette score, using an **equal-n** sampling scheme (each of the three sources down-sampled to the same count before comparing, since literature so outnumbers the other two that an uncontrolled comparison would just be dominated by it) and bootstrap-resampled 100 times for a confidence interval.

k	Query source	Neighbour source	Mean fraction	95% CI
10	literature	literature	0.316	[0.283, 0.350]
10	literature	miviludes	0.307	[0.273, 0.338]
10	literature	interviews	0.378	[0.347, 0.413]
10	miviludes	literature	0.112	[0.089, 0.137]
10	miviludes	miviludes	0.674	[0.641, 0.720]
10	miviludes	interviews	0.213	[0.183, 0.241]
10	interviews	literature	0.057	[0.043, 0.074]
10	interviews	miviludes	0.088	[0.072, 0.104]
10	interviews	interviews	0.855	[0.837, 0.875]

Table 3: k-NN neighbour composition (equal-n, $k = 10$): of a query point's 10 nearest neighbours, what fraction come from each source, on average. Rows sum to 1 within each query source.

Reading this table in plain terms: interview points are the most "self-clustering" of the three – 85.5% of an interview point's nearest neighbours are other interview points. MIVILUDES points are also fairly self-clustering (67.4%). Literature is the least self-clustering of the three: only 31.6% of a literature point's nearest neighbours are other literature points, with the rest split fairly evenly toward MIVILUDES (30.7%) and interviews (37.8%) – meaning literature, despite being the largest and most "central" source in raw point count, does not occupy a tightly separate region; it sits closer to a shared middle ground than either MIVILUDES or the interviews do relative to it. The $k = 20$ results (in the same CSV) show the same pattern.

The silhouette score for the three-source grouping, same equal-n sampling, is **0.019** (95% CI [0.017, 0.022]) – very close to zero. In plain terms: treating "literature / MIVILUDES / interviews" as three geometric clusters does not match the actual shape of the data well at all. The three sources overlap substantially rather than forming three separate blobs – consistent with, though not the same measurement as, the mixed k-NN composition above.

4.4 Criterion neighbours: what sits near MIVILUDES's 17 official criteria?

Module: `analyze_criterion_neighbours`

For each of MIVILUDES's 17 official criteria (e.g. "authoritarian, opaque group," "deceptive recruitment," "difficulty leaving the group"), this module finds which literature, MIVILUDES, and interview expressions sit closest to it in the shared space, and which of the three reference vocabularies' terms are nearest.

Because MIVILUDES's criteria are officially in French but the rest of the corpus is mostly English, this comparison is run in **three separate, never-merged ways**: (1) the criteria in their original French, projected into the main shared space (the primary representation); (2) the criteria's own stored English translations, projected through the exact same fitted transform into the identical space (the main cross-check on whether language affects the result); and (3) a simpler, secondary diagnostic comparing raw English embeddings by cosine similarity, without any of the shared-space machinery (not directly comparable in scale to the other two).

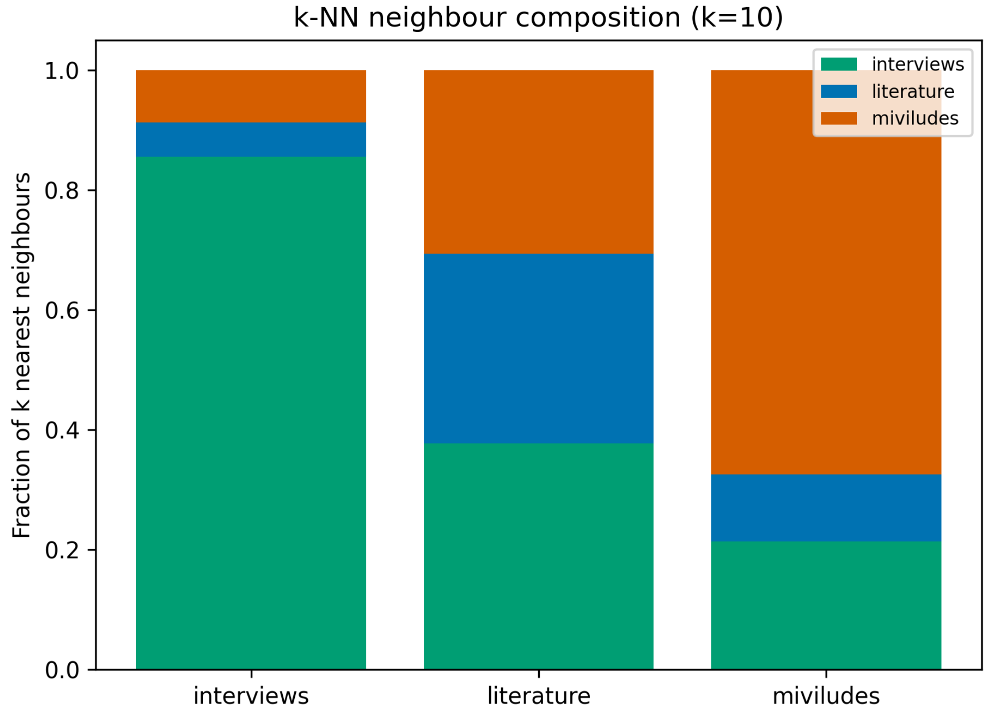

Figure 2: k-NN neighbour composition at $k = 10$, equal-n sampling, visualised.

Language sensitivity result. Comparing which corpus is "nearest" to each of the 17 criteria under the French-original versus English-translation representations: **0 out of 17** criteria change which corpus is nearest. Comparing against the simpler raw-cosine diagnostic: **1 out of 17** criteria differ. In plain terms: the choice of French vs. English representation, within the main shared space, essentially does not change which corpus each criterion is closest to.

A focused, readable 2-D projection is also generated per criterion, showing just that criterion and its nearest neighbours rather than the whole 44,000-point space. As one example:

4.5 Interview initial exemplars: what do people say first?

Module: `analyze_initial_exemplars`

This is where the manually-reviewed interview-prototype layer (Stage 6, Section 2's point-role discussion) is actually used. For each of the 26 interviews, the participant's verified first response to the opening prompt was placed in the shared space and its distance measured to: each of the 17 MIVILUDES criteria, each reference vocabulary, each expression-corpus centroid, and the nearest literature / MIVILUDES / other-interview expressions.

Each response was also given a descriptive code, `exemplar_type` – not a claim about a natural category, just a label for what kind of answer it was (e.g. a named group, a "classic" religious-NRM description, a digital/technology reference, something metaphorical, or simply unclear).

Concretely, the 26 opening responses ranged from named groups ("Ku Klux Klan," "Illuminati," "Charles Manson – and CIA") through classic religious-NRM descriptions ("smaller church denominations with a charismatic leader and their own little special lore") to metaphor-

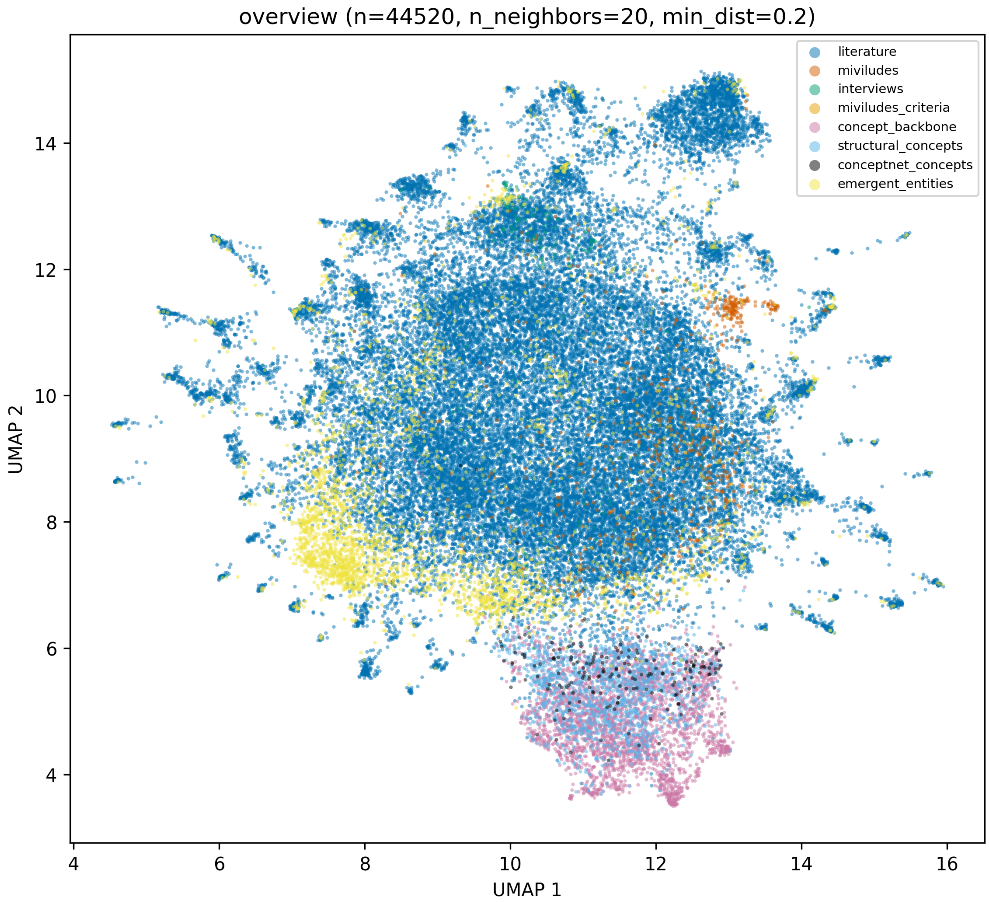

Figure 3: Whole-space UMAP overview, coloured by source dataset. At over 44,000 points this is a dense, exploratory picture – useful for a general sense of overlap and shape, not for reading off any one specific comparison.

Exemplar type	Count
unclear	11
classic_nrm_or_religious_group	4
named_group_or_movement	3
metaphorical_or_other	2
mixed	2
digital_or_technology	1
organisation_or_institution	1
unavailable (no qualifying claim in transcript)	1

Table 4: Distribution of the 26 interviews' opening responses by descriptive type. "Unclear" (11/26) is the largest single group – these are responses that describe a feeling or a vague scenario rather than naming a specific group or a textbook definition (e.g. "Something suspicious, something dangerous, something not normal").

ical/other uses entirely disconnected from the religious sense ("it's a local community of people growing local tomatoes," "cult films are usually films that were not very economically successful"). This spread is itself informative about how loosely "cult" gets used in everyday speech, independent of any geometric measurement.

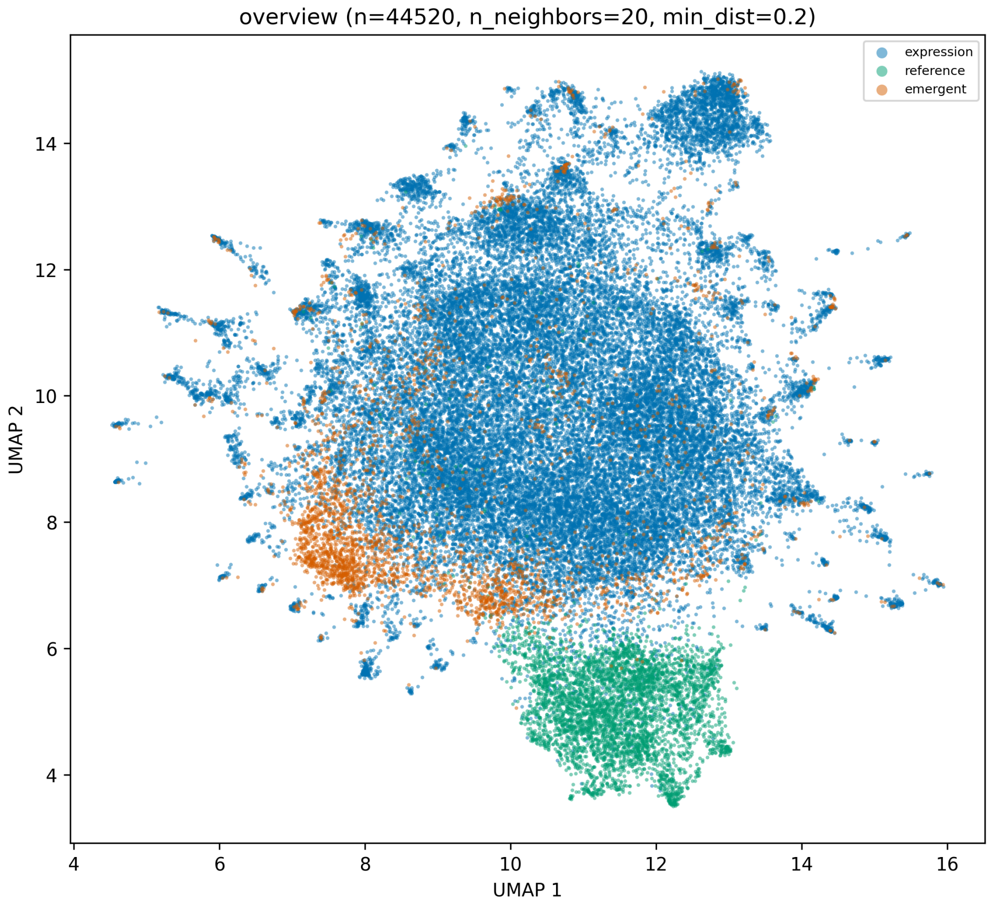

Figure 4: The same whole-space UMAP layout, coloured by point role (expression / reference / emergent) instead of source dataset.

Scope of this evidence. 25 of 26 interviews resolved into a usable geometric position; one had no qualifying opening claim anywhere in its transcript at all (a genuine absence, not a filtering artefact). This is exploratory, descriptive evidence from a small convenience sample – not a representative survey and not a basis for statistical inference. No claim of representativeness is made or should be read into the numbers here.

4.6 Emergent entities: what named things does the corpus actually talk about?

Module: `analyze_emergent_entities`

This module looks at the 3,251 emergent entities (Stage 4) and asks a simple provenance question for each one: *which* of the three corpora mentions it?

In plain terms: the overwhelming majority of named entities (3,111 of 3,251, about 96%) are mentioned only in the literature – unsurprising given literature contributes 97% of all expression points to begin with, and a direct consequence of corpus size rather than, by itself, evidence about what different sources "really" focus on. Only 9 entities are mentioned across all three corpora at once, and just 2 are mentioned exclusively by the interviews and nowhere else. This module deliberately does **not** classify any entity as "genuinely a cult" versus "mainstream" or apply any other normative label – it reports who talks about what,

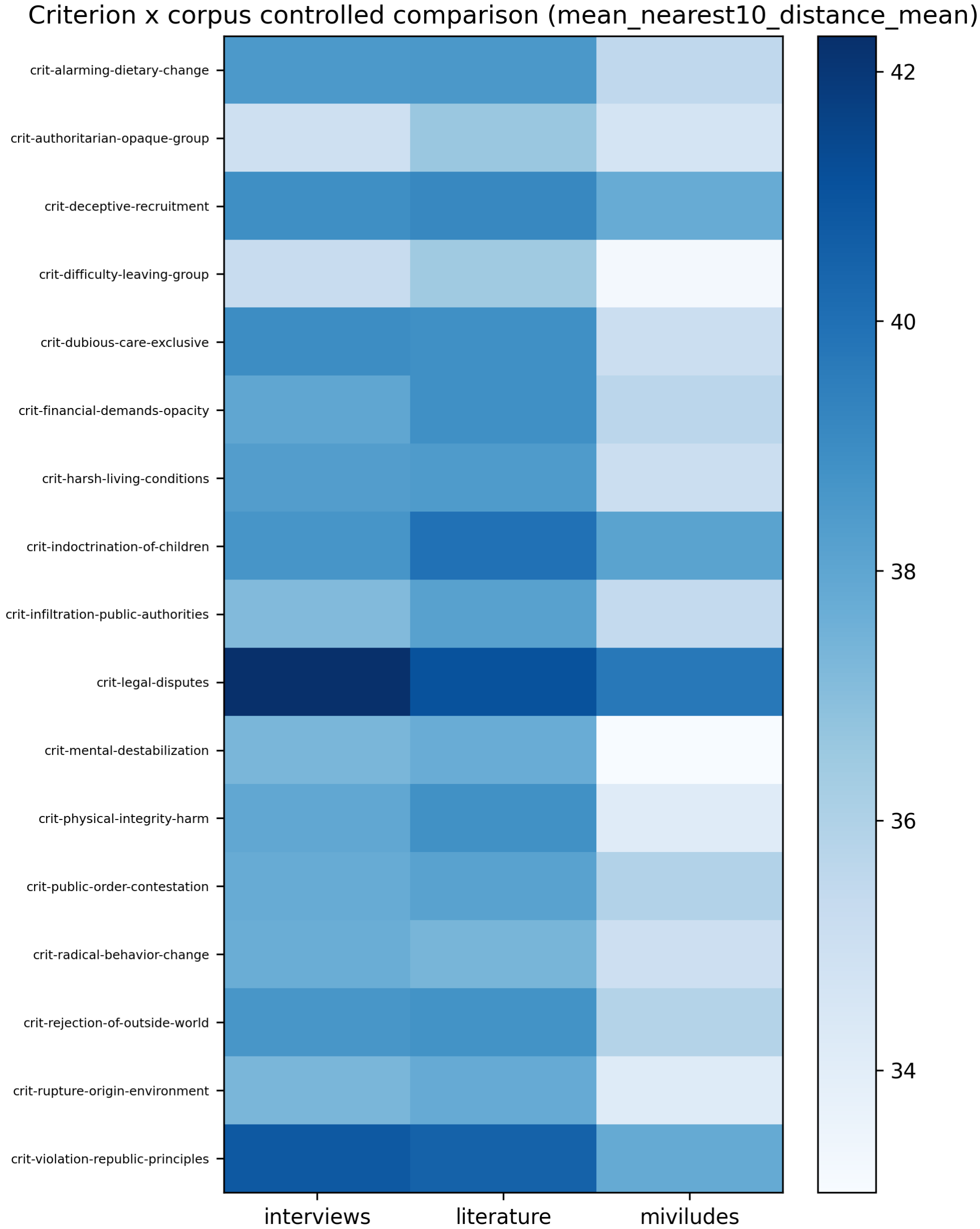

Figure 5: Heatmap of each of the 17 MIVILUDES criteria against the three expression corpora (French-primary shared-space representation).

Provenance category	Number of entities
literature_only	3,111
shared_2_corpora	81
miviludes_only	48
shared_all_3	9
interviews_only	2

Table 5: Which corpora mention each emergent entity.

not who is right.

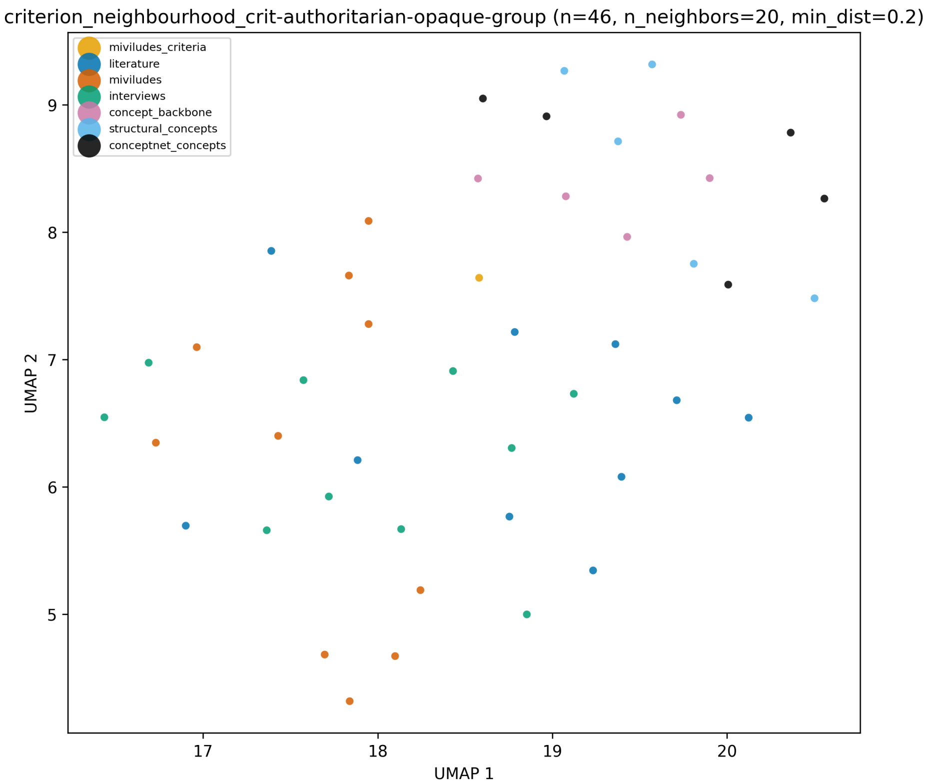

Figure 6: Focused UMAP neighbourhood around the "authoritarian, opaque group" criterion, coloured by source dataset. One of 17 such per-criterion figures generated this run; the rest are in the run's own `generate_figures/` directory.

4.7 Focused projections: readable, curated pictures

Module: `generate_focused_projections`

The whole-space UMAP pictures shown above are useful for a general sense of shape, but with over 44,000 points they are too dense to read any one specific comparison off of directly. This module instead builds small, curated point-sets around a specific question – e.g. "just the neighbourhood around one MIVILUDES criterion," "just the interview prototypes against everything else," "just the emergent-entity region" – and projects each one on its own, both via a fresh UMAP fit and via the same fixed linear PCA basis used everywhere else (so PCA figures stay comparable to each other specifically). 22 such focused populations were generated this run, ranging from 46 points (a single criterion's neighbourhood) up to several thousand (the three big reference-vocabulary comparisons). Every rendered figure notes how many points were actually fit versus how many were only overlaid afterward (e.g. corpus centroids, added to the picture after the projection was already computed, never part of the fit itself) – see the run's `generate_focused_projections/config.json` for the full list of populations and their exact sizes.

5 Honesty notes – what these numbers do not show

This section collects, in one place, the caveats already scattered through the sections above, since they matter for how any of this gets written up.

- **Corpus imbalance.** Literature is about 97% of all expression points. Any unweighted statistic over the pooled space is dominated by literature's own vocabulary and register

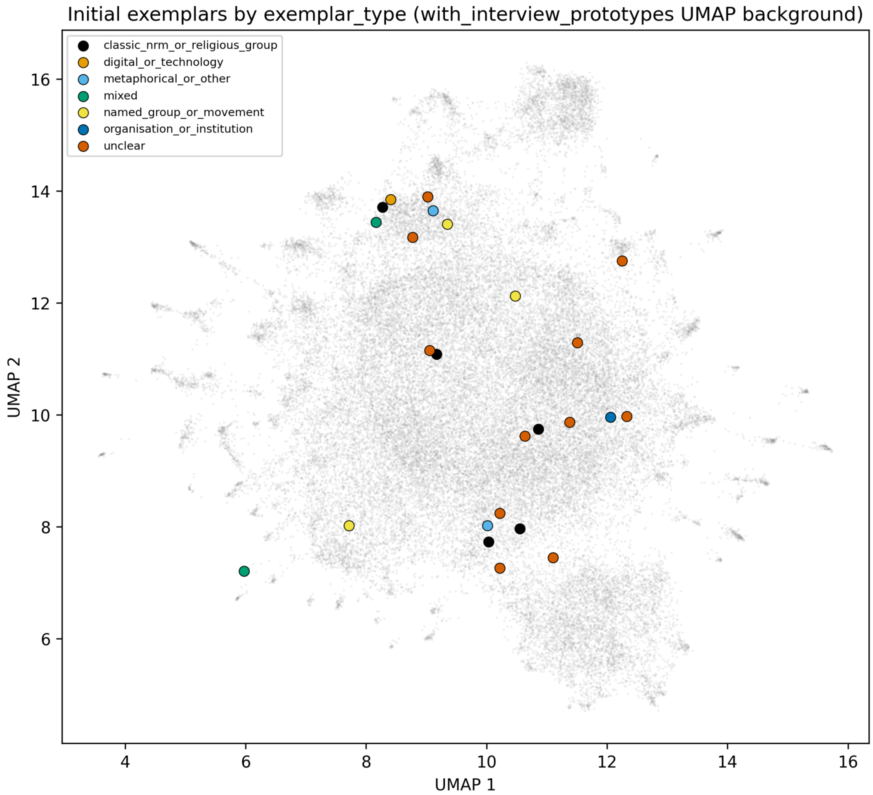

Figure 7: Interview opening responses grouped by exemplar type.

unless explicitly corrected for (as the equal-n cluster-structure numbers above are).

- **MIVILUDES is two documents.** Treat MIVILUDES's results as one influential operational framework, not a representative sample of French state framing in general.
- **The interview sample is a convenience sample.** Recruited through the researcher's own network – exploratory prototype evidence, not a representative survey of lay usage.
- **`response_rank` is not usable as a "first thing they thought of" signal** (Section 4, first subsection) – only the manually-reviewed initial-exemplar layer should be used for interview-side "first impression" claims.
- **Equal-size-controlled comparisons control for pool size only.** They are not evidence that the three reference vocabularies are otherwise interchangeable – they still differ in source, curation method, and coverage.
- **2-D pictures (UMAP, PCA slices) are a reading aid, not evidence.** The actual distance and similarity numbers come from the full 394-dimensional space; a 2-D projection can visually compress or distort real structure.
- **Three language representations of the MIVILUDES criteria are kept separate on purpose** and never averaged together, since they answer a "does this hold regardless of language" question rather than a "what is the true number" question.

6 Using this document to write the Results chapter

Everything above is deliberately descriptive: what was measured, what came out, and what it does and does not show. It does not yet answer the thesis's actual question – whether the three

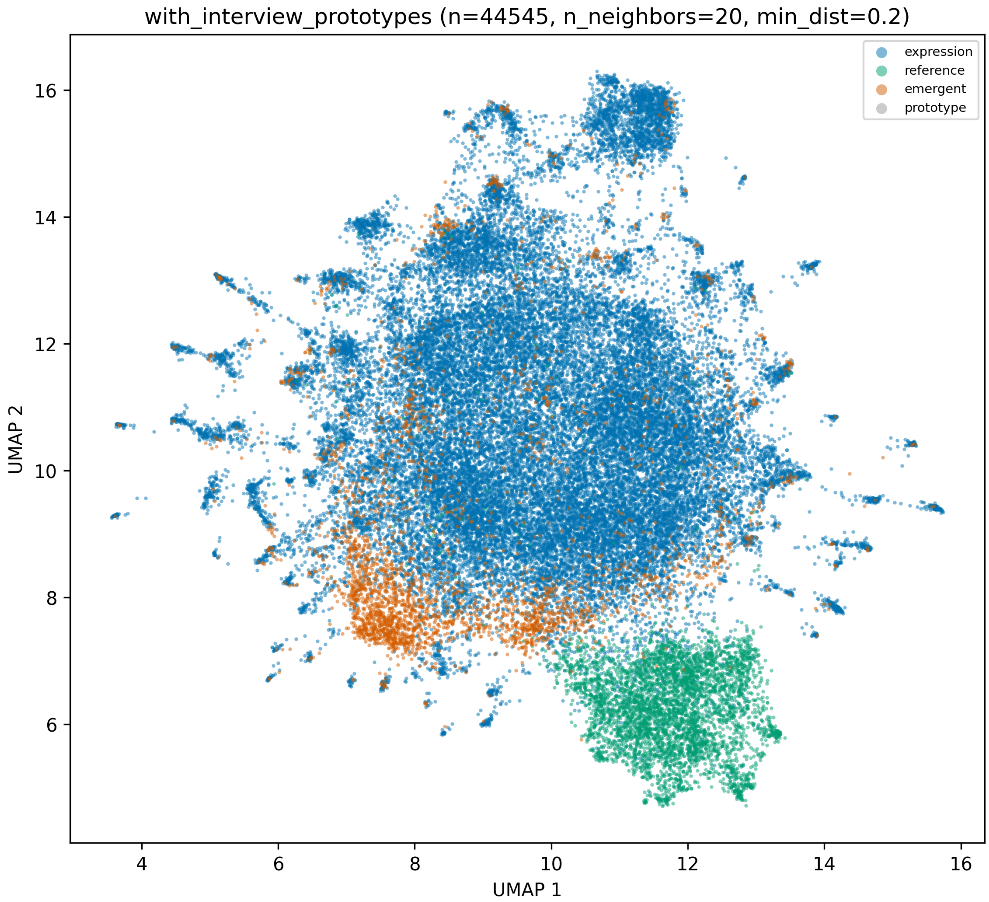

Figure 8: Whole-space UMAP with the 25 resolved interview-prototype points overlaid (point role distinguishes them from ordinary expression points), coloured by point role.

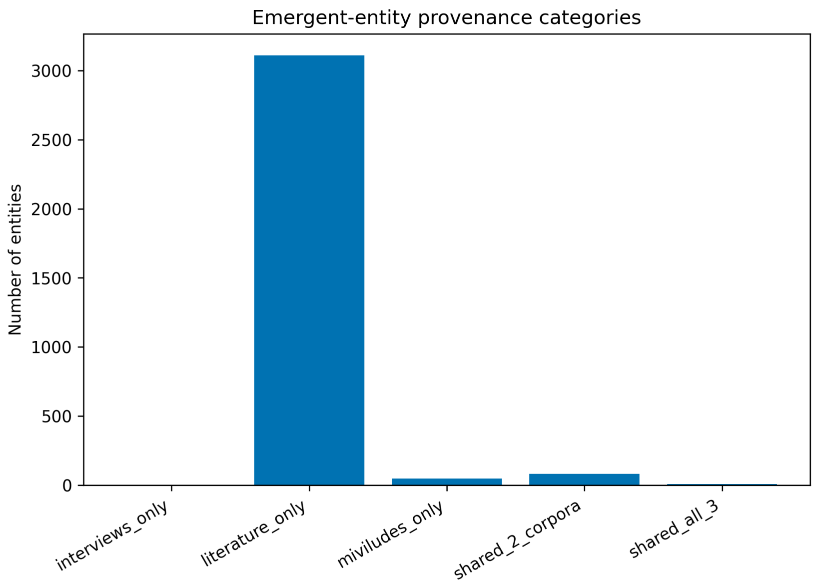

Figure 9: Emergent entities grouped by which corpus/corpora mention them.

sources converge on recurring criteria, whether "cult" fragments into genuinely different things across academic, institutional, and everyday discourse, or what kind of category (religious group, social relation, form of authority, risk category, . . .) each source implicitly treats it as. That interpretive step is the point of the Results chapter itself, and is intentionally left undone here.

Concretely, as raw material for that chapter, this document supplies: a plain-language glossary that can be trimmed down and folded into the Methods or Results chapter as needed; the actual current numbers for global structure, cluster structure, criterion neighbours, interview exemplars, and emergent entities; a curated set of figures (the ones embedded here, plus roughly 140 more per-criterion and per-population figures in `processed/analysis/20260907-002832/generate_figures/` for anything not covered by the ones shown above); and an explicit list of what these numbers do not license concluding, so that whatever gets written next stays as epistemically careful as the toolkit's own draft report already is.